NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL, RUMOMASI

TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: SEVEN

CLASS: SS3 SUBJECT: Geography

TOPIC: Geographical Data & Techniques

Introduction

- Geographical data are the facts a geographer works with — physical (rainfall, temperature, river flow), human (population, trade) and cartographic (maps and air photos). Data may be primary (collected in the field) or secondary (from books, the internet, offices).

Collection techniques

- Random, systematic and stratified sampling; questionnaires and interviews; observation and the counting of traffic, people, shops; field measurements (tape, compass, GPS); meteorological readings from the school station; and the use of maps, satellite images and statistics from the NBS, FAO and the World Bank.

Presentation

- **Tables** — the data in rows and columns; **bar charts** — to compare quantities; **pie charts** — proportions of a whole; **line graphs** — trends over time; **histograms** — frequency distributions; **flow-line and dot maps** — movement and density; **isopleth maps** — joined equal values (isohyets for rainfall, isotherms for temperature); and **distribution maps**.

Analysis and interpretation

- Calculate means, ranges, percentages and densities; describe the pattern (trend, concentration, spread); state the relationships (correlation between rainfall and vegetation); and draw conclusions with evidence. A good answer reads the data and says what it means.

CLASS EVALUATION

1. Define geographical data.
2. Name two sampling methods.
3. Which chart shows proportions of a whole?
4. What is an isohyet?

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL, RUMOMASI

TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: EIGHT

CLASS: SS3 SUBJECT: Geography

TOPIC: Surveying & Cartography

Introduction

- Surveying is the measurement of the positions and heights of points on the earth's surface; cartography is the art and science of map making. Together they produce the maps we read in geography.

Simple surveying

- **Chain/tape surveying** — measuring distances and right angles with a chain, tape, ranging poles and cross-staff to draw a plan; **plane tabling** — drawing while sighting with an alidade; **compass traversing** — bearings with a theodolite or prismatic compass; **levelling** — differences in height with a level and staff; **GPS surveying** — satellite positioning (now the standard).

Map elements

- Scale (statement, R.F., linear), grid (latitudes and longitudes), graticule, contours, conventional signs (legend), north arrow, and the title. Map projections (Mercator for navigation, Peters for equal area) distort the globe to a flat sheet.

Map reading calculations

- Scale conversion; distance and area by grid; gradient = vertical interval ÷ horizontal distance; bearing from north; and the interpretation of relief, drainage and land use from contours and the legend.

CLASS EVALUATION

1. Define surveying.
2. Name four instruments used in chain surveying.
3. What is a legend on a map?
4. Calculate the gradient between two points 2 km apart with a height difference of 120 m.

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL, RUMOMASI

TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: NINE

CLASS: SS3 SUBJECT: Geography

TOPIC: Development & Planning

Introduction

- Development geography studies the differences in wealth and welfare between regions and countries — what causes under-development and how planning can reduce it.

Measures of development

- **Economic** — GDP/GNP per capita, the share of industry, infrastructure; **social** — literacy, life expectancy, infant mortality, access to health care (the Human Development Index combines income, education and health); **environmental** — sustainability, the depletion of resources. Nigeria's HDI has been rising slowly but remains low relative to its income (oil).

Theories and issues

- The north–south divide; the dependency theory (the periphery exports primary products to the core); the role of colonisation, the oil economy and the Dutch Disease; urban bias; the brain drain; and the Sustainable Development Goals (SDGs — the 17 goals and 2030 agenda) as the global framework.

Planning for development

- Regional planning (the balance between the centre and the periphery); national development plans; the provision of infrastructure (energy, transport, water); investment in education and health; industrialisation and diversification; agricultural modernisation; and environmental protection.

CLASS EVALUATION

1. Define development.
2. Name three measures of development.
3. What is the Human Development Index?
4. State two ways Nigeria can accelerate development.

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL, RUMOMASI

TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TEN

CLASS: SS3 SUBJECT: Geography

TOPIC: Fieldwork & Practical (Project)

Introduction

- Fieldwork is the practical study of geography outside the classroom — a survey, count or investigation with a written report. It trains observation, measurement and the interpretation of the real world.

Planning fieldwork

- Choose a topic with a clear aim (e.g. 'Traffic flow at the main market junction', 'Waste disposal in my school'); state the hypothesis; select the area and method (counting, measuring, questionnaires); prepare the instruments and recording sheets; and obtain permission.

Collecting and presenting the data

- Work safely in groups; record accurately (tally counts, times, interviews, photographs, GPS points); present your data in tables and simple charts; and note the problems you met (weather, bias, missing data).

The report

- Title, aim and hypothesis, location (with a sketch map), the method, the data (tables and graphs), the analysis (the patterns found), the conclusion (was the hypothesis true?), the limitations, and recommendations. A good report is honest, ordered and illustrated.

CLASS EVALUATION

1. Define fieldwork.
2. State the steps of a fieldwork project.
3. Give two methods of collecting primary data.
4. What is a hypothesis and why is it stated?

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL, RUMOMASI

TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: ELEVEN

CLASS: SS3 SUBJECT: Geography

TOPIC: Nigeria's Population & Settlement

Introduction

- Population geography studies the size, growth, distribution and composition of the people — and the settlements they live in. Nigeria is Africa's most populous country, and its population is both a resource and a challenge.

Population size and structure

- Nigeria's population is above 200 million (the UN estimate, with the census set to confirm) — the sixth largest in the world; the growth rate is about 2.4% a year (over 3 million more people a year). Structure: a young population — about 45% is under 15 (a broad-based population pyramid); the dependency ratio is heavy (about 75 dependants per 100 workers).

Distribution and migration

- The population is unevenly distributed: dense around Lagos, the Southeast and the northern cities of Kano and Kaduna; sparse in the middle belt, the forests of the coast and the semi-arid north-east. Migration: rural–urban migration (the push of poor farm incomes and the pull of the cities) has swelled Lagos, Kano, Ibadan, Port Harcourt and Abuja; international migration — the diaspora of professionals to Europe and America (the brain drain).

Population problems and policy

- Problems: pressure on schools, hospitals, housing and jobs; unemployment; congestion and slums; pressure on land (fragmentation) and food; and the strain on infrastructure. Policy: family planning, education, the census, the creation of jobs, the development of the rural areas to reduce the drift to the cities, and the harnessing of the youth dividend.

Settlements

- Rural settlements (villages, hamlets — farming and primary activities) and urban settlements (towns and cities — commerce, administration, industry); site (where — a river, a route, a hill) and situation (surroundings); functions; and the urban problems — congestion, waste, housing and the informal economy.

CLASS EVALUATION

1. Define population structure.
2. What is the dependency ratio?
3. State two causes of rural-urban migration.
4. Mention three population problems in Nigeria.
5. Differentiate a site from a situation.

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL, RUMOMASI

TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TWELVE

CLASS: SS3 SUBJECT: Geography

TOPIC: Environmental Problems & Conservation

Introduction

- The environment is the sum of the physical and biological surroundings of man. Human activity degrades it — and the degradation returns to harm human life. Conservation is the wise use and protection of the environment for present and future generations.

The problems

- **Deforestation** — the felling of forests for fuel and farmland (Nigeria loses about 3.5% of its forest a year); **desertification** — the advance of the Sahara in the north (the 'Great Green Wall'); **soil erosion** — gully erosion in the east (Agulu–Nanka), sheet erosion, coastal erosion (Lagos, Bayelsa); **flooding** — from heavy rains and blocked drains (the 2012 and 2022 floods); **pollution** — oil spillage in the Niger Delta, gas flaring, air pollution in the cities, the pollution of rivers, waste and plastics; **urban blight** and **climate change** — rising temperatures, irregular rainfall.

Causes and effects

- Causes: population pressure, poverty, poor farming (bush burning, overgrazing, farming on slopes), logging, oil and mining operations, poor waste disposal, weak enforcement. Effects: the loss of soil fertility and farm land, the loss of biodiversity, erosion and siltation of rivers, the destruction of homes and roads, health hazards, and the worsening of food insecurity.

Conservation measures

- **Soil** — contour ploughing, terracing, mulching, cover crops, crop rotation, the planting of trees (agroforestry), check dams (the gully reclamation works); **forests** — afforestation and reforestation, forest reserves and national parks (Yankari, Cross River, Okomu, Gashaka), the ban on illegal logging; **water** — the treatment of effluent, the control of oil spills (the Ogoniland clean-up), the regulation of gas flaring; **waste** — recycling, the plastic-ban policies, proper dumpsites; **institutions** — NESREA (environmental standards and enforcement) and the Federal Ministry of Environment; **climate** — the national climate change policy, clean energy, the Great Green Wall.

CLASS EVALUATION

1. Define conservation.

2. Name four environmental problems in Nigeria.
3. State three soil conservation measures.
4. What is desertification and how is it fought?
5. What is the role of NESREA?